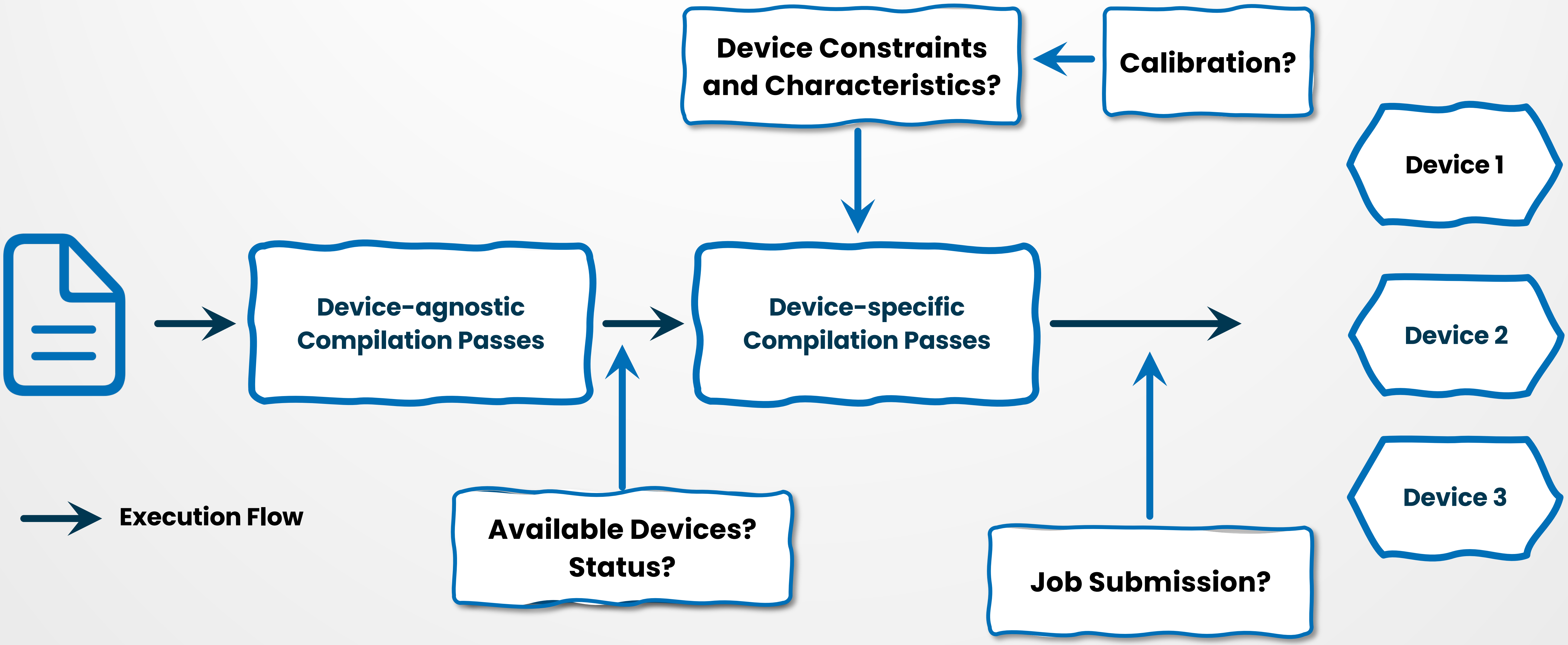
https://mq.sc

QDMI
Quantum Device Management Interface

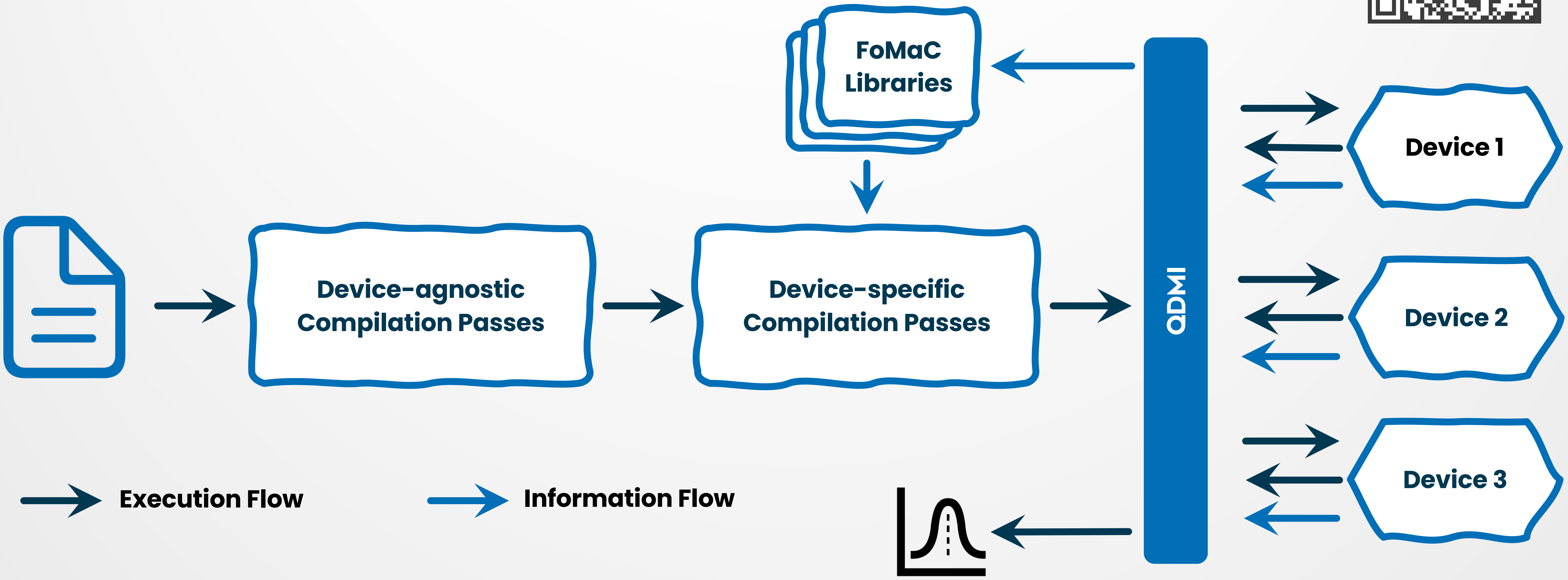
Open Source

MQSC

QDMI (Quantum Device Management Interface)



QDMI (Quantum Device Management Interface)



QDMI (Quantum Device Management Interface) – Facts

A *consumes*,
B *implements* the interface

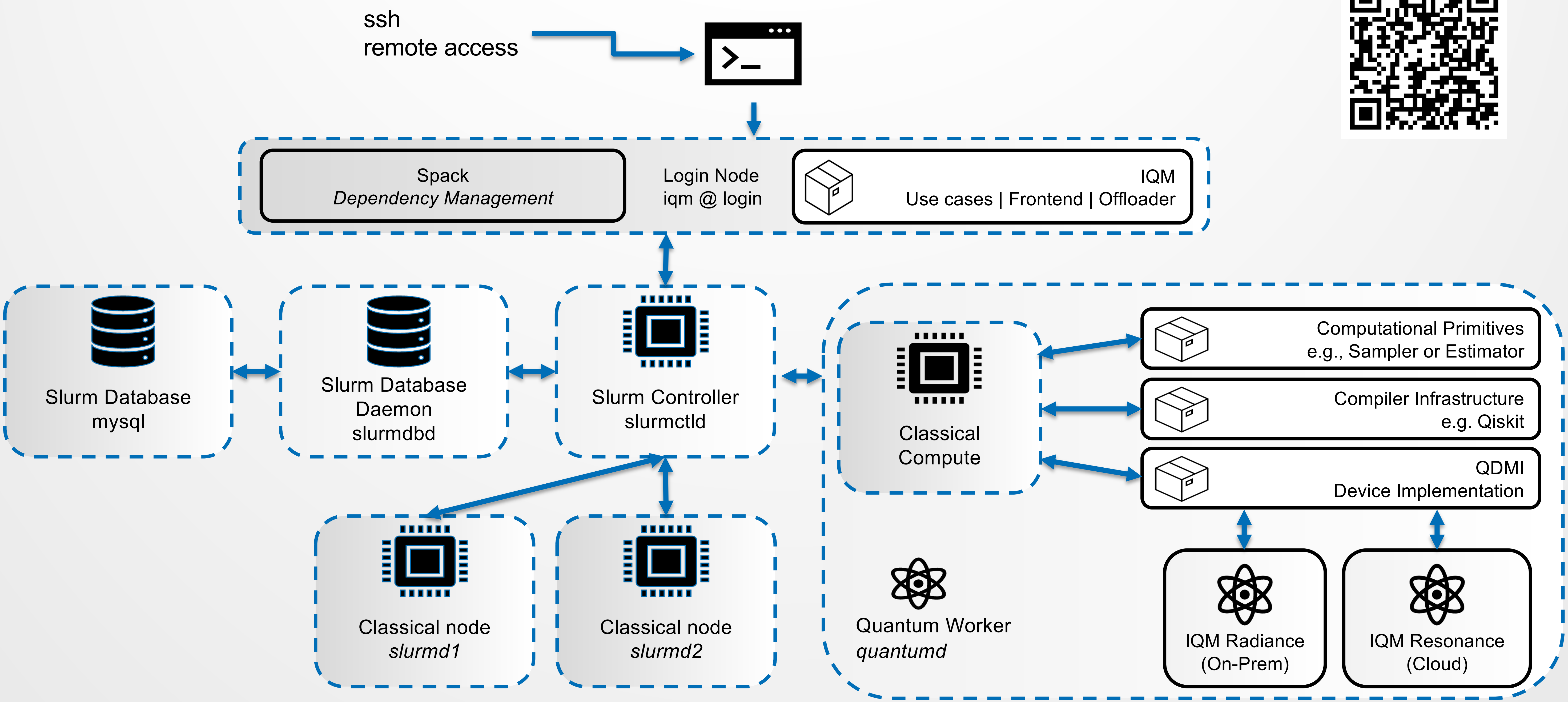
Session
- User Management
- Access Control
- Resource Management

Query
- Device Properties
- Site Properties
- Operation Properties

Job
- Job Configuration
- Job Submission
- Result Retrieval

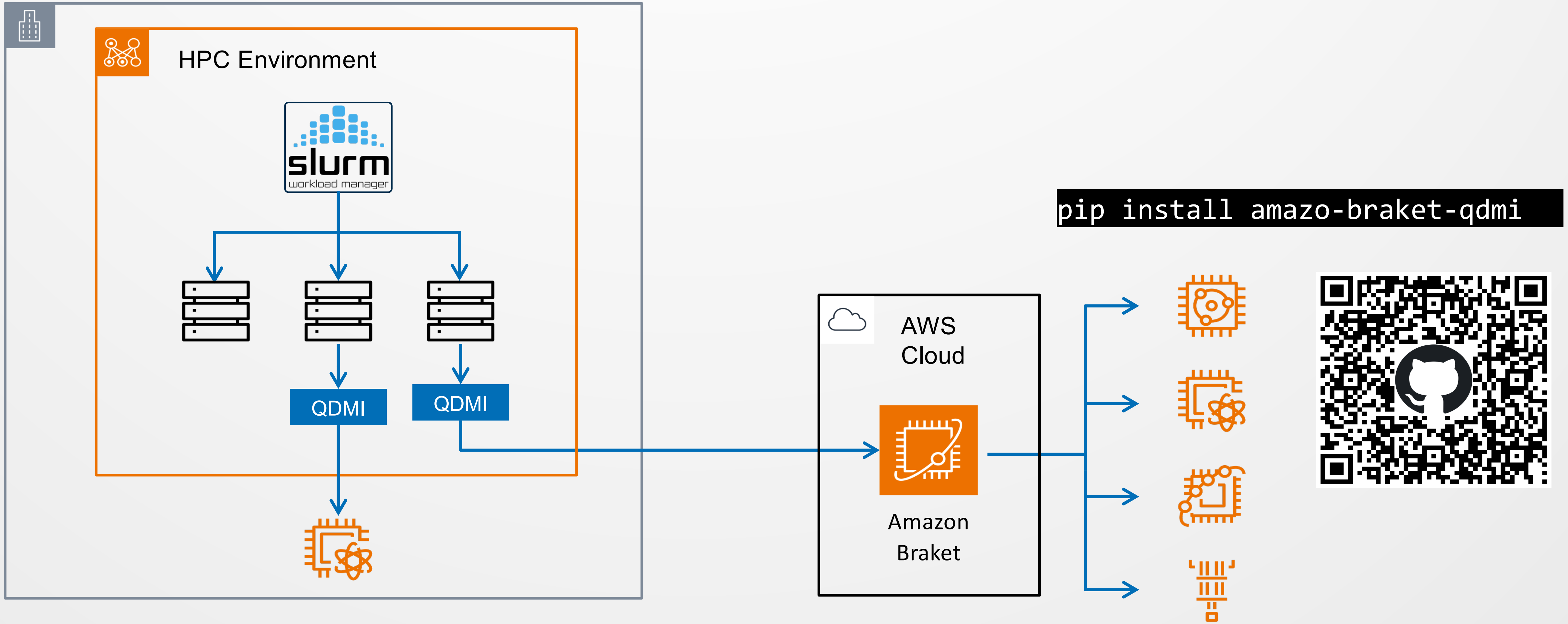
- C – level interface
- Device provider implements a dynamic/ shared library
- Typed device properties with explicit schemas
- End-to-end demonstration Qiskit to IQM via QDMI and SLURM
- Used as interface in multiple HPCs: LRZ, PSNC, CESGA, CINECA, ORNL

QDMI – IQM & SLURM Integration



QDMI (Quantum Device Management Interface) – Future Plans

- Tighter integration with compiler infrastructure
- Connecting more Hardware vendors and SDKs (Pennylane, CUDA – Q, Guppy,…)
- More/ direct low – level support for modalities (currently superconducting and neutral atoms)
- Connecting/ integrating cloud services (e.g. AWS)



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QDMIv2
Future Plans - WIP

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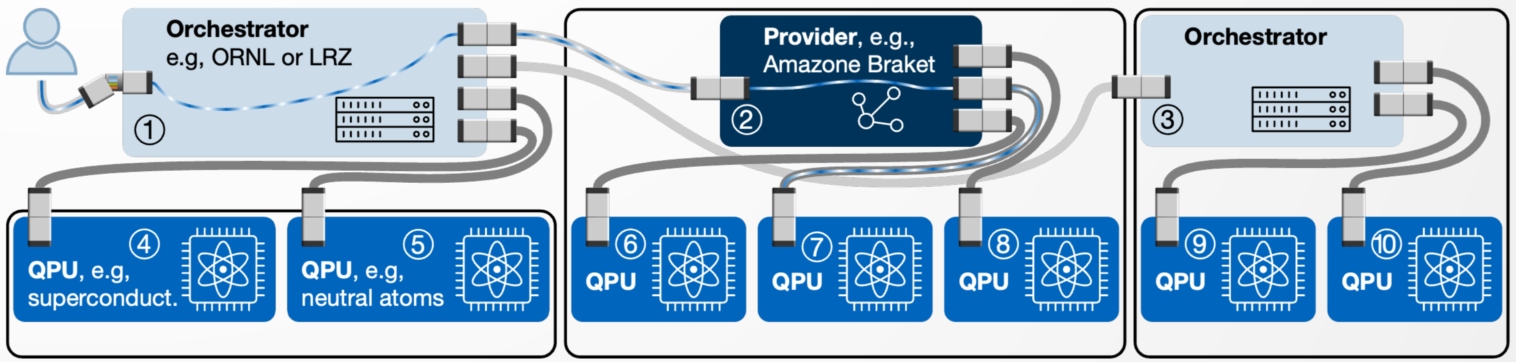
QDMIv2 – Leaner Modularity

HPCQC needs more diversity in two directions:

- ➢ Technology: Different hardware modalities
- ➢ Deployment: QPUs are accessed in diverse ways



Goal:

Have an interface which is flexible enough, while also not being fractured.

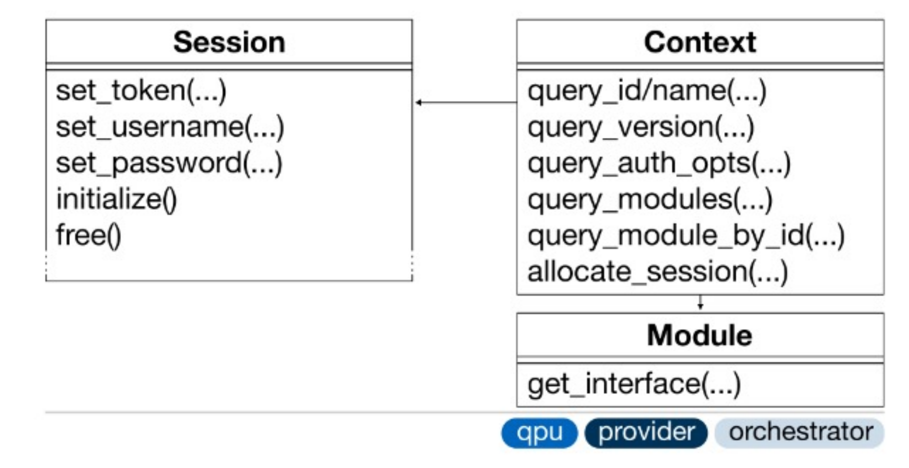
QDMIv2 – Leaner Modularity

QDMIv2: Core interface

Context

Session

Module



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QDMIv2: Extension Modules

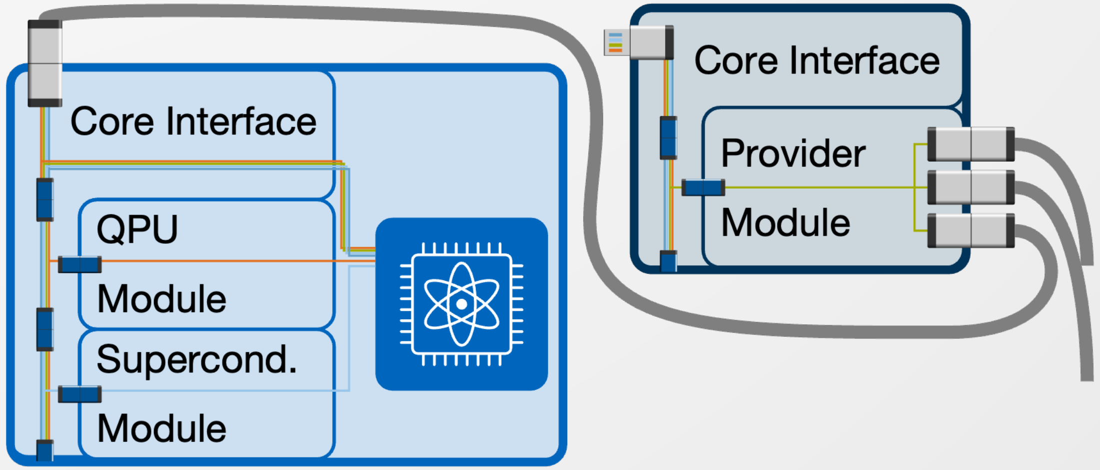
Technology

QPU

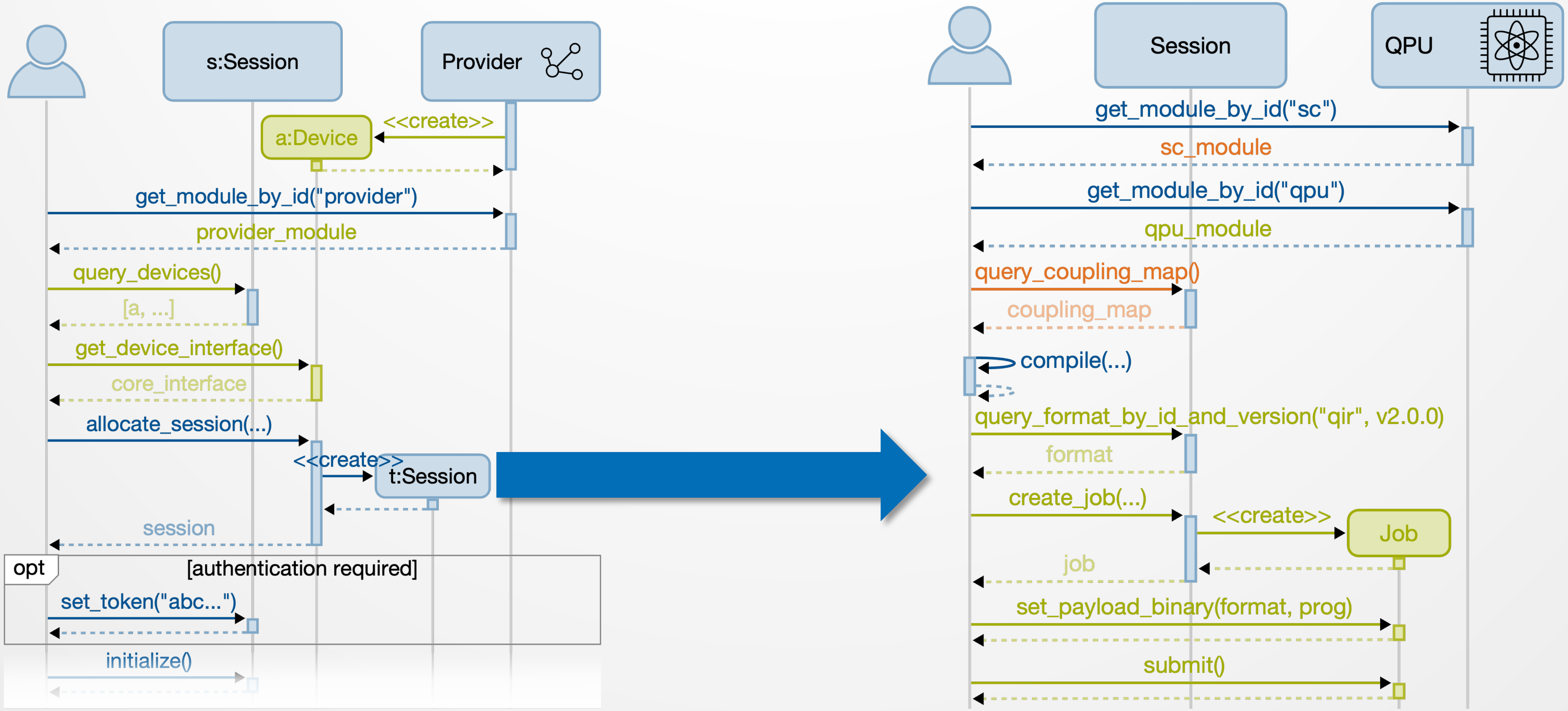
Orchestrator

Provider

Custom



QDMIv2 – Workflow from Provider to QPU



QDMI
Summary

QDMI – Summary

- C interface designed for low overhead access to quantum hardware
- Two – way communication: Job submission, result retrieval and property query
- Connected to SLURM (Spank plugin)
- Used by LRZ, PSNC, CESGA, CINECA, ORNL
- QDMIv2 with improved modularity and extensibility